



Stem cell therapy in Alzheimer's disease: possible benefits and limiting drawbacks

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Abstract

Alzheimer's disease (AD) is the sixth leading cause of death globally and the main reason for dementia in elderly people. AD is a long-term and progressive neurodegenerative disorder that steadily worsens memory and communicating skills eventually leads to a disabled person of performing simple daily tasks. Unfortunately, numerous clinical trials exploring new therapeutic drugs have encountered disappointing outcomes in terms of improved cognitive performance since they are not capable of halting or stimulating the regeneration of already-damaged neural cells, and merely provide symptomatic relief. Therefore, a deeper understanding of the mechanism of action of stem cell may contribute to the development of novel and effective therapies. The revolutionary discovery of stem cells has cast a new hope for the development of disease-modifying treatments for AD, in terms of their potency in the replenishment of lost cells via differentiating towards specific lineages, stimulating in situ neurogenesis, and delivering the therapeutic agents to the brain. Herein, firstly, we explore the pathophysiology of AD. Next, we summarize the most recent preclinical stem cell reports designed for AD treatment, their benefits and outcomes according to cell type. We briefly review relevant clinical trials and their potential clinical applications in order to find a unique solution to effectively relieve the patients' pain.

Keywords Alzheimer's disease · Neurodegenerative disorders · Stem cell therapy

Introduction

Alzheimer's disease (AD) is a devastating neuropsychiatric disorder in elderly people, inevitably associated with irreversible and progressive cognitive decline and memory loss changes in sensory-motor functions. AD is regarded as a vital concern to modern communities if no effective treatment

developed properly in the upcoming years, undoubtedly owing to the significant rise in the numbers of affected individuals and the huge medical care expenses [127]. In 2015, approximately 29.8 million people have been globally affected by AD, estimated to reach up to 115 million in 2050 [55]. The underlying neuropathological conversions in AD are recognized by chronic neuroinflammation and neuronal loss due to extracellular senile plaques (SPs), composed of amyloid- β peptide (A β) deposits, intra-neuronal accumulation of aberrantly hyperphosphorylated tau protein forming neurofibrillary tangles (NFTs), accompanied by an extensive variety of metabolic dysfunctions. The deposition of SPs and NFTs have been postulated to be the causative factors to the collapse and cholinergic neural death, initiating from the hippocampus region to the other parts of the brain, resulting in the progressive erosion of memory and gradual deterioration in cognitive ability [4]. Based on this fundamental concept, during the two previous decades, there has been a drastic increase in clinical trials meticulously designed upon the clearance of A β plaques or hyperphosphorylated tau by either immunotherapies or therapeutic agents with

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the desire of ceasing or even reversing disease progression; yet there has been no remarkable accomplishment in clinical translation of these possible strategies. Currently, hormone and cell-based therapies have gained immense attention as feasible alternatives to conventional treatments. Although immunotherapy merely provides some proper neuroprotection impacts by eliminating or blocking SPs and NFTs, it is not capable of reversing the progression of disease or replacing degenerated neurons and also there is an overwhelming risk of developing autoimmune disease [68, 108]. In addition, AD is a complex disease affecting multiple cell types and cell procedures like oxidative stress, mitochondrial dysfunction, hormone imbalance, inflammation, and calcium fluctuation, which means targets other than removing toxic protein deposits, should be regarded.

Therefore, developing novel and effective therapeutic approaches that better ameliorate disease pathologies in terms of removing toxic deposits, replacing lost neurons,

stimulating neural precursors, impeding neural death, enhancing structural neuroplasticity, and providing better circumstances for the remained cells is highly required. Recent breakthroughs with stem cells in the pre-clinical investigation and clinical trials have sparked the light of hope in treating intractable neurodegenerative diseases like AD. Stem cells possess an exceptional ability to self-renew, proliferate, differentiate, and reprogrammed for multi-lineage commitment [69]. Owing to these properties, they are currently regarded as the most appropriate choice for providing uniform distinctive cells required for cell replacement therapy. In general, there are two major identified stem cell types: (1) pluripotent stem cells comprising of embryonic stem cells (ESCs) and induced pluripotent stem cells (iPSCs), and (2) adult stem cells (ASCs) which include neural stem cells (NSCs), hematopoietic stem cells (HSC), mesenchymal (MSCs) and olfactory ensheathing cells (OECs) (Fig. 1).

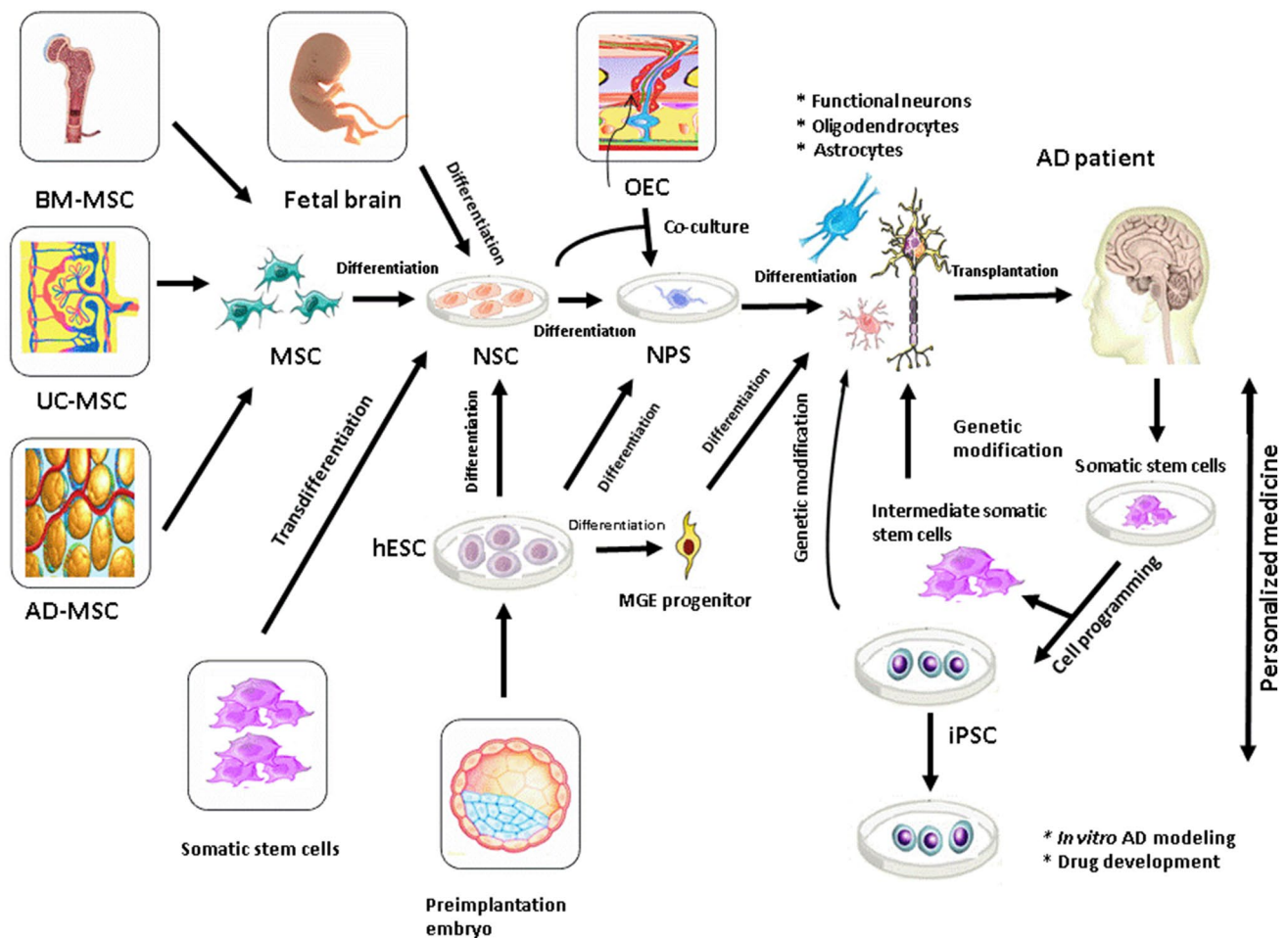


Fig. 1 Various stem cell resources applied for the treatment of AD. AD Alzheimer's disease, AD-MSC Adipose tissue-derived mesenchymal stem cells, BM-MSC Bone marrow-derived mesenchymal stem cells, hESC Human embryonic stem cells, iPSC Induced pluripotent

stem cells, MGE progenitor Medial ganglionic eminence-like progenitor cells, MSC Mesenchymal stem cells, NSC Neural stem cells, NPC Neural progenitor cells, OEC Olfactory ensheathing cells, UC-MSC Umbilical cord mesenchymal stem cells

Different types of stem cells applied for the treatment of AD

Embryonic stem cells

ESCs obtained from the inner cell mass of blastocysts, are pluripotent as they are capable of self-renew unlimitedly and develop three primary germ layers (ectoderm, mesoderm, and endoderm) [137]. ESCs an excellent candidate for cell replacement therapy approaches, provided that their pluripotency is precisely regulated into the required neural lines [102, 125]. A number of factors, including sonic hedgehog (Shh), retinoic acid (RA), and leukemia inhibitory factor (LIF) facilitate differentiation of ESCs into cholinergic neuron cells [89]. However, utilizing RA has been proved to induce direct differentiation of human ESCs into basal forebrain cholinergic neurons (BFCNs) [11].

Several reports have explored the role of ESCs and their derivatives in mitigating AD pathology in rodent models. ESC-derived neural progenitor and primed neural progenitor cells transplanted into the hippocampus of ibotenic acid-lesioned rat model effectively differentiated into functional cholinergic neuron cells, increased synapse density, and thus displayed a prominent improvement in memory deficits with no evidence of tumoral proliferation, indicating that *in vitro* differentiation of ESCs into specific NSC cell lines can circumvent tumor creation and also improve the safety and also efficacy of administration [89]. Degeneration of BFCNs and synapses throughout the basal forebrain, particularly in the nucleus basalis of Meynert (NBM), causes a pathological cholinergic insufficiency, which in turn, leads to a steady cognitive decline in AD patients. Therefore, repopulating degenerated BFCNs would be a significant step towards ameliorating the cognitive symptoms connected with AD. ESC-derived BFCNs, obtained from either mouse or human ESCs, differentiated into mature cholinergic neurons and restored cognitive performances following their injection in murine hippocampal slice cultures or into the brain of AD transgenic mice [11, 137]. Some studies demonstrated that the administration ESC-derived medial ganglionic eminence (MGE)-like progenitor cells generated BFCNs that functionally restored cholinergic innervation in the basal forebrain of AD mouse model and repaired cognitive impairments [80, 122]. In the mouse model of NBM lesion, displaying a substantial disruption of the working memory, the transplantation of mouse ESC-derived neuron-like cells (NLC) illustrated ameliorating impacts on working memory due to the development of choline acetyltransferase positive (ChAT⁺) neurons and their migration to the cortical cortex [54]. Nevertheless, some

controversial issues like immune rejection, ethical concerns, and the risk of teratoma formation upon transplantation have precluded the clinical translation of ESC [51]. Certainly, further study on the tumorigenic possibility of ESCs, and so other pluripotent stem cells demonstrated that the presence of undifferentiated pluripotent cells as contaminants in fully neural differentiated transplants can be eliminated by employing some novel technologies such as transferring suicidal genes into undifferentiated stem cells [65].

Induced pluripotent stem cells

Reprogramming fully differentiated somatic cells into iPSCs, by utilizing defined reprogramming factors, has provided a novel strategy to produce patient-specific pluripotent cells suitable for autologous transplantation [104]. Attempts to stimulate endogenous neurogenesis, substitute the lost neuron or reverse pathological changes by iPSCs modulation have confirmed early effectiveness in some AD models. In a PDAPP transgenic mouse model, the human iPSC-derived cholinergic neuronal precursor administration was shown to stimulate endogenous neurogenesis and reverse the spatial memory impairment [38]. Human iPSC-derived macrophage-like cells genetically modified to express neprilysin-2, A β -degrading protease, differentiated into functional neurons and therapeutically diminished A β levels after administration to a 5XFAD transgenic AD mouse model [118]. Furthermore, the inoculation of human iPSC-NSCs into the hippocampus of stroke mice model was shown to significantly improve neurological function. A probable explanation of these improvements could be a shift from a pro- to anti-inflammatory cytokine reaction through neurotrophin-associated reprogramming effects [31]. In a more recent report, employing protein-induced iPSCs, apotransferrin released by mESCs substantially facilitated the maturation of oligodendrocyte differentiation, and thus decreased plaque deposits and ameliorated the cognitive dysfunction once transplanted bilaterally to the brain of 5XFAD transgenic AD mice, highlighting the significance of the stem cell niche in promoting the differentiation of transplanted cells towards a distinct line [21]. In spite of these successful investigations, autologous iPSCs may exhibit genetic instability, phenotypic neuropathology, such as a marked rate of A β burdens, shortened neurite length, and increased tau phosphorylation which hinders its applications in AD clinical setting [8, 91, 121]. Nevertheless, the defective genes can be removed from the unfavorable autologous iPSCs by implementing gene editing technologies like recombinant homologous, transcription activator-like effector nucleases (TALENs), clustered regularly interspaced short palindromic repeats (CRISPR-cas9), and subsequently, healthy neurons can be transplanted into AD subjects [14]. Moreover,

to date, the high efficiency and reproducibility of the automated iPSCs reprogramming procedures have resulted in the generation of stable, high-quality lines either with main disease modeling or cell therapy. Inspiring studies imply that neuronal subtypes, for instance, cortical pyramidal and BFCNs are effectively produced [101].

Adult stem cells

Neural stem cells/neural progenitor cell

NSCs are self-renewing, multipotent cells that can differentiate into neurons and other cells of the neuronal network of the CNS. NSCs predominantly reside in the hippocampal dentate gyrus (DG) and subventricular zone of the ventricles (SVZ) of the adult brain. It is broadly accepted that the prime rate of neurogenesis, a well-established procedure that constantly generates new neurons, substantially shrinks by normal aging and these circumstances can be exacerbated by progressive and extensive cell death in AD patients [36, 147]. Since adult hippocampal neurogenesis plays a crucial role in learning and memorizing new things, optimizing this process could assist in confronting the amnesic indications of the early AD. Remarkably, endogenous NSCs are typically tended to be affected and upregulated through the extrinsic administration of chemical agents and growth factors including: allopregnanolone [23], fluoxetine [58, 106], Erythropoietin [20, 50, 144], granulocyte colony stimulating factor (G-CSF), AMD3100 and stromal cell-derived factor-1 α (SDF-1 α) [112], brain-derived neurotrophic factor (BDNF) [12, 73, 103], insulin growth factor-1 (IGF-1) [85, 95] nerve growth factor (NGF) [17, 71, 73, 128], vascular endothelial growth factor (VEGF) [52], growth and differentiation factor 11 (GDF11) [111], and transforming growth factor β (TGF- β) [136] in AD animal model.

Mounting findings signify the transplantation of NSCs and adult-derived neural progenitor can potentially ameliorate the amnesic signs and symptoms of AD via substituting for lost or damaged cells [42, 122], stimulating endogenous neural precursors and synaptogenesis [1, 140], rescuing injured inhabitant cells through producing local neurotrophic [1, 12, 89, 103, 119] and neuroprotective factors [73], transferring therapeutic genes [13, 71, 95, 128], providing immunomodulatory factors [74], decreasing neuroinflammation [143], lessening tau and A β neuropathological burdens [62, 73], and consequently improving cognitive deficits (Table 1).

Several studies have convincingly demonstrated that exogenous transplanted NSCs may differentiate into multiple neural types and properly promotes hippocampal neurogenesis predominantly through secreting some growth factors. The direct transplantation of postnatal neural stem/progenitor cells (NSPCs) into the brain of aged 3xTg-AD transgenic

mice models, depicting both A β and tau neuropathology, showed positive effects in triggering in situ neurogenesis and synaptogenesis through the secreting elevated level of BDNF vital for neural cell survival and differentiation, accompanying with preventing the uncontrolled development and progression of disease and also rescuing cognitive performance, while A β and tau loads remained intact. The role of BDNF in promoting cognitive abilities was also confirmed by applying shRNA-mediated BDNF which halted this healing process [12]. In line with this study, the NSPCs differentiated into astrocytes once transplanted into the brain of transgenic AD mice expressing human P301S tau, with progressive neuronal loss in the cerebral cortex due to astrogliosis, and promoted cortical neuronal cell survival via increasing the level of neurotrophins, specifically the glial cell-derived neurotrophic factor (GDNF) and activity-dependent neuroprotective protein [47]. Further, it was reported that intracerebrally transplanted fetal hNSCs into the NSE/APPsw transgenic mice brain, expressing APPsw under the control of neuron-specific enolase, differentiated into several neural types, precluded tau phosphorylation, and thus enhanced spatial memory. Therefore, it can be conceivably hypothesized that the high quantities of excreted neurotrophic factors BDNF, NTF3, NTF4, NGF, VEGF, FGF2, GDNF, activates Akt/GSK3 β signaling pathway thereby preventing p-tau and A β formation [73]. Indeed, overexpression of neurotrophic factors (NR2B, Trkb/BDNF, SYP, and PKC ζ) have been shown to increase the differentiation of NPCs and synaptic density, and so mitigate cognitive dysfunctions in APP/PS1 upon embryonic NSPCs transplantation [141]. The inoculation of human CNS stem cells (HuCNS-SC) into the hippocampus of treated CaM/Tet-DTA, depleted with hippocampal CA1 neurons, or 3xTg-AD mouse models efficiently restored the cognitive abilities via upregulating the expression of synaptic and growth-related markers but had no effect on A β or tau pathology [1].

Generally, the use of NSCs is impeded by the weak survival of grafted cells. Nourishing the grafted cells with neurotrophic factors has been recommended so as to raise survival of the stem cells. In Thy1-hAPP751, hAPP transgenic mice with human APP751 mutation under the control of the mouse Thy-1 promoter, administration of NSCs treated by CerebrolysinTM (CBL), a peptidergic combination having neurotrophic-like features, augmented survival of NSCs due to decreasing numbers of caspase-3 and TUNEL positive cells yet increasing furin levels which promotes the processing of pro-BDNF into BDNF in the grafted cells [103]. In addition, the administration of hNSC cells to a pretreated Tg2576 mouse with neurotrophic drugs enhanced the population of early neurons expressing doublecortin (DCX+), lessened the A β levels significantly, and in this manner delayed more cognitive decline [79].

Table 1 Selected stem cell transplantation studies of Alzheimer's disease in the last decade

Cell type/ref.	Animal model	Additional intervention	Delivery route	Obsevation	Main findings
Mouse ESC-derived NPCs and primed NPCs [89]	Drug-induced AD rat	Shh, RA and LIF signalling pathways applied to induce ESCs into cholinergic neurons	Bilateral hippocampus injection	CNLs engrafted, survived, differentiated into ChAT ⁺ cholinergic neurons, promoted synaptogenesis, and sufficiently restored cholinergic functions in transgenic mice No tumor formation in the long-term graft Improved learning and memory performance	Differentiation of stem cells toward a distinct line before transplantation is a reliable way to prevent undesired side effects of stem cell
Human BM-MSCs [66]	2xTg-AD mice	Encapsulation of human BM-MSCs bearing the GLP-1 gene	ICV injection	Reduced A β deposits Suppressed microglial and glial activation	Encapsulation of MSCs expressing GLP-1 may improve brain environmental conditions and pave the way for MSCs to exert its beneficial effects
BM-MSCs [112]	2xTg-AD mice	Treated with G-CSF/AMD3100 (CXCR4 antagonist) and human recombinant SDF-1 α	Bilateral hippocampus injection OF MSCs	Lots of BMCs promoted to migrate into the brain Microglial switched to a neuroprotective phenotype A β deposits remained intact in the hippocampus Improved learning and memory performance	Combinational therapy may enhance the recruitment of BM-MSCs into the brain
Human NSC.F3.ChAT [97]	Drug-induced AD Rat	Genetically-modified hNSCs to overexpress ChAT	ICV injection	NSCs engrafted, survived, and differentiated into immature cholinergic neurons Augmented acetylcholine neurotransmitter rate Enhanced synaptogenesis due to an increased level of BDNF Improved learning and memory performance	Genetically modified hNSCs overexpressing ChAT can nourish the injured neural cells with acetylcholine neurotransmitter and may be a promising combinational therapy to treat AD
Human AD-MSC [98]	Young and old mice	—	ICV injection	Human AD-MSC differentiated into neural cell Augmented acetylcholine neurotransmitter rate Rebuilt neuronal integrity presumably mediated by secreting growth/neurotrophic factors Improved cognitive function	AD-MSCs are a precious source of MSCs which can be used as a therapeutic agent for AD treatment

Table 1 (continued)

Cell type/ref.	Animal model	Additional intervention	Delivery route	Obsevation	Main findings
Human UCB-MSCs [133]	2xTg-AD mice	Tricyclodecan-9-yl-xanthogenate were applied to induce hUCB-MSCs into neural-like cells	Bilateral hippocampus injection	Declined brain amyloid- β deposits and enhanced soluble A β by increasing microglial activation, insulin-degrading enzyme, and neprilysin expression MSCs secrete anti-inflammatory factors like IL-4 and suppress the expression of IL-1 β and TNF- α which can activate microglia Improved learning and memory performance	Neuron-like cells derived from MSCs trigger some anti-inflammatory pathways and switch microglial to a neuroprotective phenotype, and therefore decrease the amount of A β and improve memory
Mouse T-regulatory cells educated by UCB-MSC [134]	Transgenic A β PPswe/PS1 Δ E9 mice	Educated UCB-MSC by co-transplantation with murine T-regulatory cells	Intracardiac injection	Declined brain amyloid- β deposits and enhanced soluble A β Decreased activated microglia and systemic inflammatory factors Improved learning and memory performance	Educating immune cells to combat AD pathology is a feasible and promising therapeutic strategy for AD treatment
Amniotic membrane-derived MSCs [61]	Tg2576 transgenic mice	—	Intravenous injection	Reduced pro-inflammatory and increased anti-inflammatory cytokines Switched microglial to a neuroprotective phenotype Improved spatial memory	The amniotic membrane constitutes an allogeneic source of MSCs which can be used as a potential source of MSCs for AD treatment
Human iPSC-derived neuronal precursors [39]	PDAPP transgenic mice	RA, Shh, and Noggin signaling pathways applied to induce human iPSC into neuronal precursors with cholinergic neuron phenotype	Bilateral hippocampus injection	No tumor formation in the long term graft Produced human iPSC-derived neuronal precursors migrated, differentiated into human cholinergic/GABAergic neurons and significantly improved spatial memory	The first experimental test that satisfactory proves the safety and efficacy of human iPSCs in halting the progression of AD
NSC. NEP [13]	3xTg-AD mice and Thy1-APP mice	NSCs genetically modified to overexpress neprilysin	Subiculum injection (3xTg-AD mice), Bilateral hippocampus injection (Thy1-APP mice)	Decrease A β deposits in the site of injection and its vicinity Increased synaptogenesis	Genetically modified NSCs can remove A β deposits and may be a promising combinational therapy to treat AD

Table 1 (continued)

Cell type/ref.	Animal model	Additional intervention	Delivery route	Obsevation	Main findings
Human NSCs [140]	2xTg-AD mice	–	Bilateral hippocampus injection	NSCs survived, migrated, differentiated into immature neurons and glia Induced endogenous synaptogenesis Improvements in aged transgenic mice were not related to altered A β or tau pathology	Human NSCs migrate, differentiate into neuronal lines, and enhance endogenous synaptogenesis
Mouse MSC.VEGF [41]	2xTg-AD mice	Mouse MSCs genetically modified to overexpress VEGF	Lateral ventricular injection	Improved neovascularization Reduced amyloid plaque in the hippocampus Improvement in behavioral deficiency	Genetically modified MSCs overexpressing VEGF can improve neovascularization and may be a promising combinational therapy to treat AD
NPCs [9]	Tg2576 transgenic mice	NPCs genetically modified to express IL-1 receptor antagonist	ICV injection	Enhanced neurogenesis and modulated plaque formation and microglial status	Genetically modified NSC to express IL-1 receptor antagonist can combat the harsh inflammatory conditions triggered by AD pathological proteins and augments the beneficial effects of NSCs transplantation
Encapsulated-CPECs [3]	A β -infused rats	Encapsulation of CPECs with alginate	7 microcapsules were implanted on each side of the cortex	Reduced apoptosis, gliosis, and oxidative stress Increased neurogenesis Improved long-term memory	Cell encapsulation technique can promise directing a pure population of the cell, avoid immune rejection, and provide a controlled drug delivery
NSCs [103]	Thy1-hAPP751	NSCs pretreated by cerebrolysin	Bilateral hippocampus injection	Augmented the survival of NSCs with no tumor formation	Pretreatment of stem cells with neurotrophic factors assuredly can be a potential combinational treatment for AD treatment
ESC-derived NSC [22]	Drug-induced AD Rat	ESCs induced into CNLs by NGF-PEG-PLGA-nanoparticles	Intracranial injection	CNLs engrafted, survived and differentiated into ChAT ⁺ cholinergic neurons and restored cholinergic functions in transgenic mice Improved cognitive performances not attributed to altered A β pathology	Nanomaterials can be applied to accurately differentiate stem cells into the desired specific cell types in vivo and in vitro

Table 1 (continued)

Cell type/ref.	Animal model	Additional intervention	Delivery route	Obsevation	Main findings
Mouse and human ESC-Derived BFCNs [137]	5XFAD and APP/PS1	SHH and BMP9 were applied to induce ESCs into BFCNs	Bilateral hippocampus injection	Mouse and human ESCs differentiated into BFCNs, exhibited functional properties in vitro and functionally integrated into the basal forebrain of AD mice once transplanted in vivo Improved cognitive functions	Transplantation of ESC-derived BFCNs can compensate for cholinergic insufficiency and thereby ameliorate the cognitive decline of AD mice
NPCs [94]	A β treated mice	Co-cultured with MSCs	ICV injection	Enhanced hippocampal neurogenesis via activating <i>Wnt</i> signaling pathway Improved hippocampal neurogenesis	Transplantation of NPCs co-cultured with MSCs is a promising approach for augmenting endogenous neurogenesis
Human UBC-MSCs [63]	2xTg-AD mice	Co-transplantation of hUBC-MSCs with NSCs	Multiple cisterna magna injections	Enhanced neurogenesis and synaptogenesis by the paracrine effect of GDF-15 Increased in A β deposits	Co-transplantation of NSCs with MSCs can amplify its therapeutic effects Multiple administration of MSCs can augment their beneficial impacts
Mouse NSCs [142]	2xTg-AD mice	—	Bilateral hippocampus injection	Enhanced mitochondrial biogenesis Improved cognitive functions	NSCs can promote mitochondrial biogenesis which is an early and prominent feature of AD
Human olfactory bulb neural stem/progenitor cells (OBNSC/NPCs)-hNGF [83]	Drug-induced AD mice model	Human OBNSC/NPCs genetically modified to overexpress hNGF	Bilateral hippocampus injection	Human OBNSC/NPC-hNGF survived, migrated, differentiated into immature neurons, oligodendrocytes, and astrocytes Improvement in spatial learning and memory, cognitive functions	NGF can enhance maintenance, proliferation, survival of OBNSC/NPCs, and increase their therapeutic opportunity
BM-MSCs [88]	Drug-induced AD mice model	Co-transplantation of BM-MSCs and GH with SLN	Intravenous injection of stem cell, oral administration of GH-SLNs	Increased antioxidant elements & anti-apoptotic proteins Enhanced neurotrophic factor Decreased inflammatory factors Improved cognitive functions	SLN loaded with GH exerts antioxidant effects in the brain, prolongs the life-span of MSCs and increases its efficiency
NSC.BDNF [129]	Tg 2576 mice	Genetically modified to overexpress BDNF	Bilateral hippocampus injection	Enhanced survival and engraftment of transplanted NSCs Increased synaptic integrity Improved spatial learning and novel object recognition	Genetically modified hNSCs overexpressing BDNF can enhance the neuronal viability, their differentiation, and is a potential strategy for AD treatment

Table 1 (continued)

Cell type/ref.	Animal model	Additional intervention	Delivery route	Obsevation	Main findings
Protein-based iPSCs [21]	5XFAD transgenic mice	ESC extracted protein was applied to induce mouse fibroblasts into iPSCs	Bilateral hippocampus injection	No tumor formation IPSCs-derived neurons survived, migrated, differentiated into glial cells, and also integrated into the hippocampal circuitry Declined brain amyloid- β depositsImproved learning and memory performance	Protein-iPSCs can circumvent the safety concerns linked to genetically modified iPSCs and has the potential to treat AD
Human UCB-MSCs [28]	Tg2576 mice	–	Intravenous injection	Decreased oxidative stress Promoted hippocampal neurogenesis and synaptogenesis via upregulating neuronal synaptic plasticity-related proteins levels including silent information regulator 1 and synaptophysin Brain amyloid- β deposits remained intact Improved learning and memory performance	MSCs can properly cognition of AD mice by reducing oxidative stress and stimulating hippocampal neurogenesis
Human UC-MSCs [126]	Transgenic AD mice	Treatment with Resveratrol	Intravenous injection	Enhanced hUC-MSCs engraftment in the brain The elevated rate of NPCs proliferation and neuronal survival Improved learning and memory performance	Pretreating MSCs with small molecules may positively enhance their therapeutic effects

A β Amyloid beta, *AD-MSCs* Adipose derived mesenchymal stem cells, *AMD3100* CXCR4 antagonist, *BDNF* Brain-derived neurotrophic factor, *BFCNs* Basal forebrain cholinergic neurons, *BM-MSC* Bone marrow mesenchymal stem cells, *BMP9* Bone morphogenetic protein 9, *ChAT* Choline acetyltransferase, *CNL* Cholinergic neuron-like cells, *CPECs* Choroid plexus epithelial cells, *G-CSF* Granulocyte colony-stimulating factor, *GDF-15* Growth differentiation factor 15, *ESC* Embryonic stem cell, *GH* Galantamine hydrobromide, *GLP-1* Glucagon-like peptide-1, *hNGF* Human neural growth factor, *NSC* Neural stem cell, *hNSC.F3.ChAT* Human neural stem cell expressing choline acetyltransferase, *hUCB-MSC* Human umbilical cord blood-derived mesenchymal stem cells, *ICV* Intracerebroventricular, *iPSCs* Induced pluripotent stem cells, *LIF* Leukemia inhibitory factor, *NEP* Neprilysin, *NGF-PEG-PLGA* Nerve growth factor polyethylene glycol–poly lactic acid-co-glycolic acid, *NPC* Neural progenitor cells, *NSC.NEP* Neural stem cell overexpressing neprilysin, *RA* Retinoic acid, *SDF-1 α* Stromal cell-derived factor-1 α , *SHH* Sonic hedgehog, *SLN* Solid lipid nanoparticle, *VEGF* Vascular endothelial growth factor

The impact of NSCs in enhancing endogenous synaptogenesis was found to occur when hNSCs were transplanted into the brain of APP/PS1 transgenic mice. Briefly, it was proposed that the engrafted cells resulting in significant improvement in the construction of new neural networks and synapse density, by upregulating the expression of synaptic proteins like synaptophysin (SYN) and growth-associated protein-43 (GAP-43) expression along with reinforcing the integral mitochondrial structure and thus improved cognitive performances [140].

The therapeutic efficiency of NSCs is predominantly imputed to their precise migration, site-targeted engraftment, and multi neural lineage differentiation. Thus NSCs could be exploited as useful natural vehicles for therapeutic delivery. Based on this concept NSCs can be genetically manipulated to raise their productivity differentiation or secretion of the therapeutic designed agents e.g., neurotrophins and neuroprotective growth factors, anti-inflammatory elements, neprilysin, plasmin, insulin-lowering enzyme, cathepsin B, Insulin-like growth factor 2 (IGF2) [87], matrix metalloproteinase (MMPs) [15, 56]. Research has revealed that the administration of genetically engineered mNSC-hNGFs in the cerebral cortex of the murine AD model provided resident NSCs with the enduring and steady secretion of hNGF and also increased cognitive deficits. While in the control AD model receiving naïve NSCs grafts, no cell engraftment was observed [128]. In a similar work, the transplanted hNSCs co-cultured with genetically modified fibroblasts releasing hNGF in the brain of ibotenic acid-treated AD rat model were able to differentiate into functional neurons and astrocytes, efficiently integrated, and mitigated cognitive performances thereby [71]. Research has revealed that the transplantation of F3.ChAT, human NSCs genetically manipulated to overexpress choline acetyltransferase, into the brain of rat AD model could significantly improve cognitive function [96]. The transplanted genetically engineered NSCs producing neprilysin was found to constantly release neprilysin, decrease A β deposits in the site of injection and its vicinity, intensify synaptic connectivity, and improve fundamental A β pathology in 3xTg-AD and Thy1-APP mice [13]. Indeed, treating Tg2576 transgenic mice, with the administration of NPCs over-expressing IL-1 receptor antagonist enhanced neurogenesis and modulated plaque formation and microglial status, improved cognitive performances [9]. The genetically altered NSCs expressing human insulin-like growth factor 1 (IGF-1) was shown to survive for 10 weeks, recover spatial learning impairment, and develop learning and memory ability of AD model [95]. The in vitro cultured IGF-1-overexpressing cortical neurons exhibited augmented GABAergic neuron differentiation, enhanced VEGF production, and exhibited a higher survival rate among the transplanted cells [85]. Of note, there has been an innovative trend of applying self-destructive NSCs

to tackle with the safety concerns pertinent to the application of genetically modified cells after migration to the site of injury in the human [24].

While the majority of studies focus on the transplantation of NSCs into the hippocampus, the transplanted hNSCs to the corpus callosum of the APP/PS1 mice survived, migrated, and differentiated into neural subtypes, signifying the feasibility of targeting the corpus callosum in AD cellular therapy [86].

In more recent studies, it is documented that NSCs may have immunomodulatory features. The inoculation of NSCs has also been shown to reverse the pro-inflammatory effects of activated microglia in APP/PS1 mice within 10 weeks via downregulation of inflammation-related pathways TLR4, MyD88, TRIF, p-P38, MAPK, and NF κ B [143]. Similarly, the transplanted NSCs into 12 month-old Tg2576 brains caused a reduction in A β plaque levels as evidenced by increased A β degrading factors, released anti-inflammatory cytokines, decreased microglial activation and neuroinflammation, increased synapse density, which resulted in improved cognitive defects, while failed to recover neuropathological features and cognitive dysfunctions in the 15 months of the same race, emphasizing on opting the appropriate time for cell therapy intervention [53]. These experiments indicate that NSCs could be induced for propagation and neuronal differentiation to mitigate cognitive impairments in AD.

Mesenchymal stem cells

Mesenchymal stem cells (MSCs) are adult multipotent progenitors which are able to self-renew and differentiate into various cell types once cultured under proper circumstances. MSCs can be retrieved from a variety of adult tissues and organs, including peripheral blood, bone marrow (BM-MSCs), umbilical cord (UCB-MSCs), adipose tissue (AD-MSCs), amniotic fluid, Wharton jelly, fetal liver, muscle, and lung [10]. MSCs has come into the limelight as an option for cell therapy as their high proliferative potency, anti-inflammatory properties, accessibility, in vitro large capacity propagation, secretion of a wide range of cytokines, and lack ethical concerns. Along with these mentioned features, MSCs can be administrated intravenously, which is the least invasive method and makes multiple injections feasible. The therapeutic mechanisms of MSC-related improvements in AD healing have not been entirely clarified but it is proven that MSCs gives rise to multiple neural cell types and enhance neurogenesis through the release of neurotrophic factors like VEGF, BDNF, IGF-1, NGF, and (FGF2) fibroblast growth factor 2 [26, 41, 63, 98] and ameliorate AD pathology thereby. Secreted neurotrophic factors can promote neurogenesis [28, 63, 94, 98, 107, 113], augment neuroprotection [41, 88], enhance A β clearance [63,

107, 113, 133, 146], reduce tau hyperphosphorylation [60], promote synaptogenesis [63], modify innate and adaptive immune cell responses through upregulating neuroprotective cytokines such as IL-10 while decreasing amount of pro-inflammatory cytokines TNF- α and IL-1 β [61, 70, 143], increase microglial phagocytic activity [59, 70, 72, 82, 92, 98], enhance neovascularization [41], suppress cell death induced by A β and tau [70], decrease oxidative stress and apoptosis [28], adjust autophagy pathways by activating the PI3K/Akt pathway, increasing anti-apoptotic factors and reducing caspase-3 activity [77, 113], reduce plaque size [92], and thus provide effective recovery for AD models.

Intravenously injected human AD-MSCs into the brain of AD mouse model differentiated into neural cell types, regulated microglial activity, and enhanced endogenous neurogenesis via the increased level of neurotrophic factors, in particular, BDNF, NGF, and thus developed cognitive and motor functions in an aged mouse model [98]. The administration of BM-MSCs obtained from the female rat in the tail vein of male of the same race, resulted in markedly promoted neurogenesis through upregulating the expression of nestin and ChAT⁺ cells and also reduced amyloid deposits in the hippocampus, signifying that MSCs can pass through the blood–brain barrier (BBB) and selectively reside on the injured sites in the brain from a long distance [107]. Interestingly, the multiple transplantations of hUCB-MSCs, secreting growth differentiation factor 15 (GDF-15) into the cerebrospinal fluid stimulated neurogenesis and synaptic activity, reduced β amyloid plaques, and enhanced cognitive function in AD mouse models [63].

The administration of MSCs co-cultured with NPCs into the brain of A β -infused mice overcame the inhibiting effect of aggregated toxic proteins on neuron degeneration and increased hippocampal neurogenesis in 4 weeks via upregulating *Wnt/ β -catenin* signaling pathway [94]. Similarly, in 3xTg-AD AD mice model, MSCs transplantation dramatically diminished the levels of highly toxic A β ₅₆ and conserved levels of glutamine synthetase levels, resulting in a reduction in the total A β deposition and reversed memory decline [105]. Remarkably, intravenous hUCB-MSCs did not exert regenerative functions, no engrafted cell or no positive effect on AD pathological hallmarks, in AD mouse model after only one intravenous injection, signifying that multiple administration is required for MSCs to exert their impacts [100]. Further work confirmed that the only one time administration of human ischemia-tolerant MSCs, expressing a reduced level of human leukocyte antigen-related (HLA-DR) cell surface receptor and exhibiting more distinguished migratory tendencies in comparison with MSCs grown in normal oxygen conditions, drastically decreased cerebral A β deposits and neuroinflammation in the brain of APPPS1 transgenic mice, demonstrating that optimizing the culture conditions

or pretreating MSCs with small molecules may positively enhances their therapeutic effects. Surprisingly, multiple intranasal application of MSCs extract reduced the amount of A β in the cerebral as well [49]. Taking this into account, it was proved that resveratrol, a polyphenolic antioxidant that activates various signaling molecules, boost MSCs beneficial effects in the recovering learning and memory impairment in AD mice by supporting hUC-MSCs engraftment in the brain, stimulating NPCs proliferation, accentuating elevated rates of neuronal survival by activating SIRT1 signal way [126].

Although MSCs is also highly favored stem cell-based treatment for AD, they have a limited lifespan and the majority of them die before having an impact on the injured region. Therefore, MSCs can be genetically manipulated to improve the efficiency of transplantation, and also to deliver augmented therapeutic and neurotrophic factors such as BDNF, GDNF, into the brain to support the injured neurons. Direct transplantation of genetically modified BM-MSCs overexpressing VEGF into the brain of a double transgenic APP^{swE}/PS1 ^{Δ E9} mouse model contributed to neovascularization, diminished A β deposits in the hippocampus of the brain, and thus improved the cognitive deficits [41]. Similar to other neurodegenerative diseases, neuroinflammation has been identified as a major participant in AD. Also, several studies have proposed further evidence that hMSC transplantation modulates some inflammatory pathways like nuclear factor kappa-light-chain-enhancer of activated B cells signaling, cell-to-cell interaction with target cells, modulation of antigen-presenting cells and sphingosine-1-phosphate signaling [81, 84, 125].

The transplantation of HUMSCs, human umbilical cord MSC-derived NLC, in a transgenic APP/PS1 mouse model modulated neuroinflammation, decreased A β deposits and consequently retrieved cognitive performance [133]. Similarly, the administration of MSCs in A β _{1–42}-injected mice was found to modulate neuroinflammation via activating glial cells, suppressing of iNOS and COX-2 expression, preventing inflammatory cytokine release, halting neuronal cell death, stimulating neuronal differentiation, and thus reduced cognitive deficits [138]. T-regulatory cells educated by MSCs effectively rescued cognitive deterioration and induced reduction in A β accumulation due to moderating microglia activation and systemic inflammatory factors in a 3xTg-AD transgenic mice models [134].

Transplantation of human amniotic mesenchymal stem cells (hAM-MSCs), as an attractive substitute supply of MSCs, into APP/PS1 mouse model activated glial cells, reduced microglial activation, and neuroinflammation, intensified release of degrading enzymes, and thus significantly reduced A β . Furthermore, hAM-MSCs promoted BDNF secretion, which in turn, augmented hippocampal neurogenesis in the SGZ of the DG and improved synaptic

integrity, resulting in retrieving cognitive ability, proposing hAM-MSCs may be a novel and effective source of MSCs supply for the treatment of AD [146].

Several studies verified that the transplantation of hMSCs restored BBB disruption, caused by neuroinflammation permitting neutrophil infiltration, and also induced morphogenic alterations in astrocytes, endothelial cells, and microglia leading to a diminished neutrophil infiltration and improved neuronal survival in AD model [46, 99].

Together, these studies provide an optimistic outlook for applying hMSCs to amend AD pathology. However, there are two sides to every coin. The dark side of applying MSCs as a therapeutic agent is their homing inclination towards hypoxic areas around tumor margins and their great revascularization capacity, which promotes the survival of probable tumors once it is solely administrated. Of date, amounting studies have concentrated on the powerful therapeutic effects of stem cell-derived extracellular vesicles. Extracellular vesicles (EVs), membrane vesicles that are secreted by a variety of mammalian cell types, have been shown to deliver bio-active products to the nearby diseased or injured cells, promoting immune modulation, angiogenesis, neurogenesis and synaptogenesis [19, 48, 67]. The first study explored the putative therapeutic utilization of MSC-EVs was in brain injuries. Systemic administration of MSC-EVs was demonstrated to stimulate neurite outgrowth in a rat stroke model. This impact is perceived as being mediated via the release of miR-133b, which can regulate gene expression interfere in neurite rearrangement in neurons and astrocytes, from MSCs-EVs to neurons, proposing a mechanism for the therapeutic advantage of MSCs [130, 131]. EVs echo the genetic profile of their parent's cell to recipient cells to modulate physiological or pathological procedures. Therefore, MSCs can be genetically altered to release EVs supplemented with therapeutic agents like neprilysin, insulin-lowering enzyme, cathepsin B, MMPs, and small interfering RNA (siRNAs) that target β - and γ -secretase enzymes to the brain. Based on this concept, the administrated engineered exosomes, acquired from dendritic cells loaded with siRNA molecules for glyceraldehyde-3-phosphate dehydrogenase (GAPDH) mediated β -site APP-cleaving enzyme 1 (BACE-1) in the brain of a rat model. Of note, EV-mediated siRNA, targeting the mRNA of β -secretase, knocked down the GAPDH gene, downregulated the brain BACE1 protein, and reduced A β levels by half in the animals [5]. NEP-enriched EVs released from AD-MSCs being co-cultured with a genetically modified neuroblastoma cell line, overproducing human A β , mediated clearance of both extracellular and intracellular A β from the brain. Significantly, neprilysin secretion was more pronounced in AD-MSCs compared to BM-MSCs, proposing the therapeutic significance of ADSC-derived exosomes for AD treatment [57]. Likewise, EVs released from dental pulp stem cells (DPSCs) were found to degrade A β ₁₋₄₂

and diminished A β -induced toxicity in neuroblastoma cells (SH-SY5Y) in vitro. Surprisingly, the expression of NEP in DPSCs was meaningfully higher as compared to in BM-MSCs and AD-MSCs, introducing a powerful source of stem cell to combat the detrimental impacts of neurodegenerative diseases [2]. Also, MSC-EVs can be used as an approach to induce neural differentiation. The delivered exogenous miR-124 mediated by MSC-derived exosomes considerably reduced the expression of its target mRNA, and actively participating in promoting the neuronal differentiation of the NPCs [45]. Based on this proof of concept studies, utilizing genetically modified MSC-EVs to either transfer therapeutic drug, release a cocktail of growth factors and siRNAs may open a new horizon for prospective strategy in AD and other neurodegenerative disease treatment. In addition, systemic injection of exosome-like liposomes (Combined exosome and liposomes by employing electroporation) containing BACE1 siRNA spread in the brain, maximized the features of the exosome surface so as to reduce its immunogenicity while enhancing its colloidal viability and boosting the half-life of exosomes in the blood [110]. In one study, it was reported that MSC-EVs suppressed inducible nitric oxide synthase (iNOS) mRNA and the level of toxic proteins induced by A β in primary cultured neurons treated with A β or obtained from APP/PS1 mice, attenuated impairments in synaptic plasticity, and long-term potentiation which subsequently mitigated cognitive decline in APP/PS1 mice [124].

Olfactory ensheathing cells

OECs are a source of multipotent stem cells present in the lamina propria that generally offer structural, metabolic and trophic support to neurons through secreting neurotrophic growth factors, NGF, BDNF, along with ECM molecules like fibronectin. Therefore, OECs provide a synergistic impact once transplanted with other stem cells. Several attempts have verified the inducement of axonal regeneration through applying co-transplantation of OECs with NPC or other neural cell lines in animal models and lately in humans. Co-culture of BM-MSCs with OECs enhanced neural differentiation in vitro [93]. Additionally, positive impacts yielded in axonal regrowth of a peripheral nerve lesion model following transplantation of co-cultured OECs derived from the olfactory bulb and the olfactory mucosa [44].

Genetically modified olfactory bulb neural stem/progenitor cells (OBNSC/NPCs) expressing hNGF also demonstrated ameliorating effects on the cognitive decline in ibotenic acid-induced lesions. Human OBNSC/NPCs administrated to CA1 region of the hippocampus were able to survive, migrate to other CA regions, proliferate, integrate and differentiate into mature neurons, oligodendrocytes, and astrocytes [83]. The beneficial effects of OECs

on suppressing A β -induced toxicity were determined by the incubation of neuroblastoma SH-SY5Y cells with A β_{25-35} , and OEC-conditioned medium (OECM). OECM normalized the cell survival via reducing apoptosis and decreasing intracellular reactive oxygen species (ROS). Nevertheless, OECM had no effect on intracellular Ca²⁺ concentration in SH-SY5Y cells, suggesting that OECM improves oxidative destruction triggered by A β_{25-35} in SH-SY5Y cells by blocking the mitochondrial intrinsic pathway [37].

Practical implications of biomaterials on AD cell therapy

A need for advanced therapeutic choices

It is anticipated that next generation therapeutic approaches will concentrate on multimodal strategies consisting of cell therapy, immunotherapy, tissue engineering, and nanotechnology to provide a more versatile treatment for AD. Regarding a hostile microenvironment as is in the brain of subjects with AD, including A β plaques and p-tau; loads of transplanted cells typically generate some non-neuronal glial cells or unexpectedly die, and thus do not specifically participate in the reconstruction of the injured sites. Moreover, there may be a spatial disparity between the injured nerve and the grafted cell [139]. In addition, the extracellular matrix of the afflicted areas does not favor stem cell survival and differentiation. Thus, 3D scaffolds can properly serve as an appropriate biomimetic platform to stimulate and support cell growth, neuron regeneration, and neuronal circuits' reconstruction in a substantially efficient manner. Although no published study based on biomaterial scaffold implantation has been performed in AD treatment yet, the potential outcomes achieved by very few investigations on other brain disorders are encouraging and to our view merit consideration for practical application in the appropriate treatment of AD [18, 32].

In vivo study has revealed that Type 1 collagen-based porous gelatinous scaffold prolonged survival of human BM-MSCs, restricted its distribution, and promoted neurite outgrowth and neural function recovery upon implantation into the core of the lesion of adult TBI rat models [43]. Additionally, another investigation demonstrated that injectable sodium alginate and hyaluronic acid scaffold, a biocompatible copolymer that encompasses cells and provides sites for cell adhesion, loaded with hUC-MSCs not only enhanced the survival rate and propagation of MSCs, but also substantially improved learning and memory impairment in the TBI mouse model via promoting the regeneration of endogenous nerve cells [145]. Besides, research has reported that the injected synthetic poly desaminotyrosyl tyrosine ethyl ester carbonate (pDTEc) polymer fiber scaffold, with tunable

hydrophobicity and degradation features, seeded with iPSC-derived neurons into hippocampal brain slices and mouse brain in vivo, efficiently stimulated in situ NSCs regeneration, synaptic remodeling, and facilitated the engraftment of neuronal cells into the brain tissue. Furthermore, the survival rate of iN cells seeded on scaffolds substantially increased by approximately 38-fold at the injection site [18]. These encouraging results highlight the beneficial interventions of scaffolding in neuronal tissue regeneration and can be effective in a clinical setting; however, it seems that injectable scaffolds seeded with stem cells tend to be more suitable for efficiently delivering a distinct cell population to the site of interest in the brain as they penetrate brain's cavities and do not disrupt the profoundly elaborate and sensitive structure of the brain.

Neural tissue engineering proposes another appealing area in which a polymeric semi-permeable, non-degrading hydrogel provides a proper niche for xenograft, allograft, or genetically altered cells to release natural bioactive substances like trophic factors or to express the desired protein for a prolonged period of time while being protected by the host immune system. Notably, contrary to conventional 3D scaffolds, encapsulated units do not thoroughly integrate into the host tissue and can simply be extracted after a while, which would minimize some safety issues and expedite the adoption of cellular therapies for clinical employment. The feasibility, safety, and long-term survival of allogeneic cells utilizing macroencapsulation have indeed been proved in some clinical setting [33, 34]. Moreover, multiple pre-clinical reports have explored the feasibility and efficacy of microencapsulation technology, coating individual cells with an immunoprotective obstacle, in AD models. For example, the administration of microencapsulated baby hamster kidney (BHK) fibroblast cells expressing VEGF resulted in a decreased brain A β load, apoptotic cell death, and hyperphosphorylated-tau expression as well as a reversal of spatial memory and learning deficits in transgenic APP/PS1 mice. Significantly, these cells amplified cellular propagation in the DG and eliminated the expression and activity of acetylcholinesterase in the diseased brain, which can be justified by the secretion of VEGF [6, 116]. In another study, the functional ability of microencapsulated human retinal pigment epithelial (RPE) cells genetically modified to sufficiently express the brain ciliary neurotrophic factor (CNTF), a neurotrophic factor which supports the survival of motoneurons, was positively determined by minimizing the toxic effects of A β oligomers and improving cognitive performances was determined in transgenic Tg2576 mice [40]. Therapeutic administration of choroid plexus epithelial cells (CPECs) has been proved as a novel approach in AD model for their intrinsic therapeutic value in releasing endogenous neurotrophic factors like NGF, VEGF, BDNF, as well as participating in regenerative activities [29, 109].

Implantation of alginate microcapsules loaded with CPECs into the brain of A β infused rats caused a significant recovery via a reduction in apoptosis, gliosis and oxidative stress and an increased in neurogenesis, resulting in improved long-term memory [3].

In the era of targeted treatment opportunities, there are also some alternative approaches that focus on optimizing stem cell therapy in AD by circumventing BBB with minimally invasive methods which is a major concern in elderly AD patients. Combining of stem cell with nanomaterials, NP-loaded and NPs conjugated to the surface of stem cells, has opened a broad spectrum of potential applications in brain regenerative studies as nanomaterials provide an ideal platform for stem cell imaging and tracking [30, 114], conveying genes to mediate stem cell proliferation and differentiation [25, 38, 120], enhancing the efficacy of stem cell treatment [88], displaying a high prosperity for guiding stem cells to the brain [16, 90, 117]. Amounting evidence suggests that some nanomaterials can induce neuronal differentiation and proliferation in stem cells in vivo circumstances. For instance, curcumin-encapsulated PLGA nanoparticles (Cur-PLGA-NPs) effectively promoted the upregulation of the genes necessary for the NSCs proliferation (Reelin, nestin, and Pax6), neuronal differentiation (neurogenin, neuroD1, neuregulin, neuroligin, and Stat3), and improved learning and memory ability in A β -treated rats through selective activation of *Wnt* signaling pathway [120]. Transplantation of rat ESC-derived NSC, induced by NGF-PEG-PLGA-nanoparticles, into the lateral ventricles of 192 IgG-saporin, treated AD rats' with cholinergic neuronal ablation dramatically developed learning and memory performance in concordance with restoring hippocampal synaptic plasticity and cholinergic neuron functions in the basal forebrain regions, while having no impact on the whole amount of A β plaques in AD mice [42]. Interestingly, self-assembling nanofiber scaffolds (SANs) which are predominantly characterized by a repetitive peptide consequence can mimic the structure of extracellular membrane (ECM) and enhance the adhesion of transplanted NSCs and axon regrowth impacts, tune neural proliferation, differentiation, neurite outgrowth and synapse development in vivo conditions [75, 78]. Moreover, this construction can be functionalized by adding specific biological motifs to support neural grafts, simulate the structure of ECM, promote the differentiation of stem cells into the desired phenotype, and also enhance the mechanical strength [27, 53, 76, 78, 115]. Notably, since some SAPs undergo the same aggregation pathways as amyloid precursor proteins do, they can be employed to intervene with APP processing and so hinder AD progress. The implantation of IKVAV-functionalized SAPs into bilateral hippocampus of APP_{sw}/PS1 _{Δ E9} transgenic AD mice model, modified the processing of APP through decreasing BACE1, whereas increased insulin-degrading enzyme and neprilysin levels leading to

a decrease in A β plaques along with the levels of soluble A β _{1–40} and A β _{1–42} in the cortex and hippocampus within 2 weeks, and thus rescued cognitive deterioration, indicating that hybrid SAPs substantially enhanced neural regeneration as compared to non-functionalized scaffolds, proposing that encapsulating stem cell with these structures can amplify their beneficial impacts [135]. In a similar manner nanoparticles small cationic SAP, taken from the main segment of A β protein, attached to poly-lactic-coglycolic acid (PLGA) nanoparticles were proven to impede A β assembling and also preserve SH-SY5Y cells from its toxic effects in vitro condition [132]. Yet in preliminary stages, these satisfactory outcomes demonstrate that transplanting SAPs with stem cells or optimizing such structures for encapsulating stem cells can magnify the therapeutic efficacy of cell therapy and are highly recommended for the treatment of neurodegenerative disorders and particularly in AD.

Clinical perspective

To date, over 400 clinical trials are exploring new treatments for AD and most of them are in progress. The majority of studies are designed for eliminating the destructive impacts of a toxic protein known as amyloid-beta and tau in the brain, but quite recently, considerable attention has been paid for the feasibility of treating AD patients with AD cell therapy (Table 2).

The past decade has witnessed the first wave of employing neural stem cell in clinical trials for AD treatment based upon the concepts that they can migrate to the degenerated regions of the brain, differentiate into multiple neuronal lineages, and also release some neurotrophic and neuroprotective factors required for the stimulation of endogenous neurogenesis.

Previously in a phase I clinical trial the transplantation of microencapsulated implants containing ARPE-19 human RPE cell lines genetically modified to express NGF (EC-NGF) to the basal forebrain of six patients with a probable diagnosis of mild to moderate AD according to the NINCDS-ADRDA (National Institute of Neurological and Communicative Disorders and Stroke-AD and Related Disorders Association) criteria showed positive results. Although the capsules were retrieved by the end of 6 months, a meaningful increase of NGF was observed in all participants, leading to a concomitant boost in the numbers of proliferating neural cells. In addition, improved cognition impairment in two patients indicated the better function of intracerebral neural networks. No adverse events were reported, proposing this procedure as a practical treatment for AD, with future improvements designed to enhance cell survival (NCT01163825) [33]. Based upon previous accomplishment, a phase 1/Ib clinical trial of the second generation of NGF-ECB genetically modified RPE cell line

Table 2 Latest clinical trials registered at *Clinicaltrials.gov* on stem cell therapy for Alzheimer disease as of September 2018

Trial no.	NCT01297218	NCT02054208	NCT02600130	NCT02672306
Type of trial	Phase I (N = 9) Single-center, open-label, safety/efficacy study	Phase I/IIa (N = 45) Randomized, quadruple-blind, parallel assignment, safety/efficacy study	Phase I (N = 30) Randomized, placebo-controlled, double-blinded, safety/efficacy study	Phase I/II (N = 40) Randomized, parallel assignment, double-blind, safety/efficacy study
Intervention	Group 1: 3 million allogeneic hUCB-MSC will be given one time using intracerebral administration Group 2: 6 million allogeneic hUCB-MSC will be given one-time using intracerebral administration Group 3: placebo to AD patient and evaluation, measurements will be carried out 12 weeks after post-administration	Stage 1: (N = 9) Group 1: 10 million cells/brain allogeneic hUCB-MSCs will be given three times at four interval weeks using ICV administration Group 2: 30 million cells/brain allogeneic hUCB-MSCs will be given three times at four interval weeks using ICV administration Group 2: placebo to AD patient and evaluation, measurements will be carried out 24 weeks after post-administration Stage 2: (N = 36) Group 1: 30 million cells/brain allogeneic hUCB-MSCs will be given one-time Group 2: placebo to AD patient and evaluation, measurements will be carried out 24 weeks after post-administration	Group 1 (N = 10): 20 million allogeneic hMSCs were given one-time using intravenous administration Group 2 (N = 10): 100 million allogeneic hMSCs were given one-time using intravenous administration Group 3 (n = 5): placebo (plasmalyte A and 1% human serum albumin) to AD patient and evaluation measurements carried out 2, 4, 13, 26, 39, and 52 weeks after post-administration	Group 1: 20 million allogeneic hUCB-MSCs will be given eight times at two interval weeks using intravenous administration Group 2: placebo to AD patient and evaluation, measurements will be carried out 10 weeks after post-administration
Main eligibility	Probable AD determined by NINCDS-ADRD and DSM IV and K-MMSE on 10–24, PIB-PET positive (SUV > 1.5)	Probable AD determined by 18–26 K-MMSE, NINCDS-ADRD, Amyloid + PIB-PET or Florbetaben PET	AD determined by NINCDS-ADRD & 18–24 on MMSE	Probable AD and mixed dementia according to NINCDS-ADRD & 3–20 MMSE
Primary outcome measures	Maximum tolerated dose and safety: Number of participants with adverse events, number of participants with a normal range of vital signs, mixed lymphocyte reaction, and laboratory tests	Safety: number of participants with adverse events	Safety of LMSCs 30 days post infusion	ADAS-Cog
Secondary outcome measures	Changes from baseline in ADAS-cog, S-IADL, K-MMSE, CGA-NPI, serum transthyretin, amyloid- β and tau in CSF, PIB-PET, and regional cerebral metabolic rate for glucose (FDG-PET) at 12 weeks post dose	Changes from baseline in S-IADL, K-MMSE, CGA-NPI, DTI mapping, CSF biomarkers, ADAS-cog response rate, PIB-PET, regional cerebral metabolic rate for glucose (FDG-PET), and CIBIC-plus	Preliminary efficacy determined by examining changes in disease status and rate of decline (at baseline, 2, 6, 13, 26, 39, and 52 weeks)	Changes from baseline in ADCS-CCGIC, MMSE, ADCS-ADL NPI, Plasma A β , CSF A β and tau
Trial status	Completed 2011.2–2012.4	Recruiting 2014.1–2019.7	Recruiting 2015.11–2018.10	Active, not recruiting 2016.1–2019.10
Outcome	After 12 weeks, no severe acute, long-term side effects, significant clinical efficacy in the evaluated measurements	Safety: 24 weeks from post-administration Efficacy: 24 weeks from post-administration	Safety: 30 days from post-administration Efficacy: 4, 13, 26, 39, and 52 weeks from post-administration	Safety: 10 weeks from post-administration Efficacy: 10 weeks from post-administration
Trial no.	NCT02833792	NCT03117738	NCT03172117	Trial no.
Type of trail	Phase IIA (N = 40) randomized, crossover study, assignment, single blind, treatment, placebo-controlled	Phase I/II (N = 60) randomized, parallel assignment, placebo-controlled, double-blinded, safety/efficacy study	Phase I/II a (N = 45) randomized, parallel assignment, quadruple-blind, follow-up study of safety/efficacy	Type of trail

Table 2 (continued)

Trial no.	NCT02833792	NCT03117738	NCT03172117	Trial no.
Intervention	Group 1: 1.5 million allogeneic hMSCs per kilogram weight body will be given one-time using intravenous administration Group 2: placebo to AD patient and evaluation, measurements will be carried out 18 months after post-administration	Group 1: unknown amount autologous hAD-MSC cells will be given nine times at two interval weeks using intravenous administration Group 2: placebo to AD patient and evaluation, measurements will be carried out 32 weeks after post-administration	Group 1: 10 million allogeneic hUCB-MSC will be given three times at four intervals using ICV administration Group 2: 30 million allogeneic hUCB-MSC will be given three times at four intervals using ICV administration Group 3: placebo to AD patient and evaluation, measurements will be carried out 24 weeks after post-administration	Intervention
Main eligibility	Mild to moderate AD determined by NINCDS-ADRD & 12–24 MMSE, NINCDS-ADRD, Amyloid ⁺ florbetapir PET	Probable mild to moderate AD determined by on NINCDS-ADRD & on 16–26 K-MMSE	Subjects completed the phase I clinical trial (NCT02054208)	Main eligibility
Primary outcomes	Incidence of adverse events	Adverse events, laboratory examination results, ADAS-Cog	ADAS-Cog	Primary outcomes
Secondary outcome measures	Neurological examinations	Changes from baseline in MMSE, CDR-SOB, NPI, GDS, ADCS-ADL, C-SSRS, MRI, CSF A β 40 & β 42, AICD, sNRG-1 biomarkers	Changes from baseline in ADAS-Cog, S-IADL, K-MMSE, CGA-NPI, CDR-SOB, CIBIC-plus, Florbetaben-PET, FDG-PET, MRI, changes in CSF biomarkers	Secondary outcome measures
Trail status	Active, not recruiting 2016.6–2018.6	Recruiting 2017.4–2018.10	Recruiting 2017.5–2021.12	Trail status
Outcome	Safety: 18 months from post-administration Efficacy: 18 months from post-administration	Safety: 32 weeks from post-administration Efficacy: 32 weeks from post-administration	Safety: 24 weeks from post-administration Efficacy: 24 weeks from post-administration	Outcome

AD Alzheimer's disease, ADAS-Cog Alzheimer's Disease Assessment Scale-Cognitive Subscale, ADCS-ADL Alzheimer's Disease Cooperative Study Activities of Daily Living, AICD Amyloid precursor protein intracellular domain, C-SSRS Columbia Suicide Severity Rating Scale, CDR-SOB Clinical Dementia Rating-Sum of Box, CGA-NPI Caregiver-administered Neuropsychiatric Inventory, CIBIC-plus clinician's interview-based impression of change-plus, CMRglc Cerebral metabolic rate for glucose, CSF Cerebrospinal fluid, DSM IV Diagnostic and Statistical Manual of Mental Disorders, DTI Diffusion tensor imaging, FDG-PET Fludeoxyglucose positron emission tomography, GDS Geriatric Depression Scale, hAD-MSCs Human adipose-derived mesenchymal stem cells, hUCB-MSCs Human umbilical cord blood-derived mesenchymal stem cells, hMSCs Human mesenchymal stem cells, ICV Intracerebroventricular, K-IADL Korean Instrumental Activities of Daily Living, MRI Magnetic resonance imaging, N Number, NGF Nerve growth factor, NINCDS-ADRD National Institute of Neurological and Communicable Disease and Stroke and Alzheimer's Disease and Related Disorders Association, NPI Neuropsychiatric inventory, PET Positron emission tomography, PIB-PET Pittsburgh compound B positron emission tomography, S-IADL Seoul-Instrumental Activities of Daily Living, sNRG-1 Soluble neuregulin-, SUV Standard uptake value

with the improved hNGF release was performed in sixteen mild to moderate AD patients. In this work, four NGF-ECB implants were transplanted in the cholinergic basal forebrain regions of the brain of patients with a possible diagnosis of mild to moderate AD according to the NINCDS-ADRDA criteria. Diagnostic tests of sixteen participated people approved that the capsules were properly fitted and after 6 month extraction, there was no proof of inflammation or capsule movement. Implanted NGF-ECB led to a meaningful increase in ChAT acetylcholine synthesizer (ChAT) and degrading enzyme (AChE) activities (compared to the AD patients of the same age) correlating with NGF release even after 6 month extraction of all the capsules. Besides, there was no evidence of a further cognitive decline in 6 months assessed by the Mini-Mental State Examination and Alzheimer Disease Assessment Scale-Cognitive subcomponent (ADAS-Cog). Nevertheless, Levels of CSF AD biomarkers A β _p and tau proteins did not alter meaningfully during the study period. After a mean follow-up of 12 months, no long-term adverse events were documented, recommending that this procedure need to be executed for an extended period of time to precisely capture positive impacts of NGF-ECB on AD pathology (ClinicalTrials.gov, Reg. No. NCT01163825) [34].

In a phase I clinical trial, the implantation of genetically engineered autologous fibroblasts programmed to produce NGF directly into the NBM mediated recovery through augmenting the function of cholinergic neurons along with an apparent positive evidence of a response to NGF with post-mortem evaluation of enrolled people, thereby demonstrating a new and potentially efficient therapy against the mild stage of AD [123].

HMSCs are numerically the most favored cell types presently under human clinical trials as they are safe and efficient, can be easily harvested and transplanted intravenously into patients, lack major ethical issues, develop minimal immunogenicity and provide a broad range of therapeutic benefit. Based upon a phase I clinical trial study conducted by Kim's research group intracranial injected allogeneic hUCB-MSCs into the bilateral hippocampi and precuneus regions administration of nine individuals probably suffering from mild-moderate AD provided no adverse events required to pass phase I, no significant clinical efficacies or neuro-protective effects were observed in AD patients based on the evaluated measurements (ClinicalTrials.gov, Reg. No. NCT01297218, NCT01696591) [64].

Currently, only two clinical trials employing allogeneic hMSCs have been initiated. One employs intravenous administration in a single-blind (ClinicalTrials.gov, Reg. No. NCT02833792), yet another utilizes intravenous infusion in a double-blind randomized, placebo-controlled approach (ClinicalTrials.gov, Reg. No. NCT02672306). Besides, four clinical trials are recruiting. A phase I/IIa clinical trial study

will use intraventricular implantation of the ommaya reservoir system containing allogeneic hUCB-MSCs in about 45 AD patients (ClinicalTrials.gov, Reg. No. NCT02054208, NCT03172117). Furthermore, a phase IIa study will examine the safety and tolerability of ischemia-tolerant allogeneic hMSCs previously cultured in hypoxic conditions when administered intravenously to the brain of AD patients (NCT02600130). And finally a Phase I/II study will explore the benefits of autologous AT-MSCs transplantation on AD pathology (NCT03117738).

Although cell therapy field expects the results of many ongoing AD clinical trials, the scientist is struggling to address some minor technical issues in this area to pave the way and to expedite developing effective therapeutics and a cure for AD. Except for insufficient investigation financing, issues with participant enrollment are certainly the most crucial obstacle to clinical investigation development. First of all, the proper time for stem cell transplantation in AD is not particularly determined yet and several Alzheimer's cell therapy trials have failed as intervention tests did not carry out at the right time. AD is a progressive chronic disease that generally initiates some years before it can be diagnosed. Consequently, by the time signs or symptoms emerge, the individual's brain damage is too advanced and significant amount of central neurons in the brain has already died, leading to an irreversible loss. Early diagnosis is impeded due to the absence of efficient biomarkers or reliable approach sensitive enough to distinguish early AD before symptoms emerge [35]. On the contrary, in some cases, brain imaging has revealed that about a few percents of the study's participants had no the amyloid pathology that the trial therapies were planning to target, suggesting pressing need to offer early detection techniques. Given that clinical trials continue for several years and patients with dementia experience several invasive injections and undergo some laborious follow-up procedures, some participants resign before the trial is finished and that makes it difficult to evaluate the outcomes. Another subtle point that unfortunately none of the clinical trials have taken into account, is the sex differences. Women are estimated to constitute approximately 2/3 of the AD population (Association, 2017) [7]. Therefore, in order to yield meaningful data that could lead to the development of an effective treatment, the randomized controlled trial, concentrating on particular populations needs to adapt and evolve to respond to the gender differences.

Concluding remarks and future perspectives

In spite of being infancy, the accelerated advancements and breakthroughs of stem cell technology positively confirm its prospective utilizations in the treatments of AD. As mentioned earlier, extensive preclinical works have been carried

out to demonstrate the capability of stem cells to treat AD. Nevertheless, further investigations are needed to elucidate the multiple practical and theoretical issues. Thus far, pre-clinical experiments have merely emphasized gathering a more thorough understanding of the mode of action of transplanted cells. Yet, it is vital to learn from these proofs-of-concepts tests; future preclinical trials should not exclusively examine safety and efficacy outcomes; preferably, they should focus on precise hypotheses on the mechanisms of efficiency. Therefore, a comprehensive evaluation is required on the source, type, dose, delivery systems, long-term safety and efficacy, the reaction of the implanted cell to the harsh pathogenic AD surroundings, and mechanisms of action in the AD model to confirm their most effective therapeutic results. Undoubtedly, the next generation strategies in AD cell therapy evolution might include a combination of multimodal strategies at distinct stages of the disease, perhaps via utilizing various genetically-modified stem cells and/or cell-free techniques, applying different approaches of tissue engineering, or other tactics to target the underlying pathological hallmarks of AD to promote neurogenesis or synaptogenesis.

Compliance with ethical standards

Conflict of interest All authors declare that they have no conflict of interest.

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